AKANSHA SUBEDI

STRUCTURAL ENGINEER

Melbourne, VIC, Australia | Australian Permanent Resident

+61495010488 | akansha.subedi@gmail.com | [LinkedIn] | [Portfolio website]

SUMMARY

Structural Engineer with 5+ years of experience in the **analysis and design of reinforced concrete and steel building structures** across residential, commercial, healthcare and institutional projects. Experienced in **structural modelling, linear and non-linear analysis, performance-based design, seismic and wind design and engineering calculations** with proficiency in **ETABS, SpaceGass, SAP2000, and Revit** and knowledge of **Australian Standards (AS3600, AS4100)**. Strong background in site inspections, multidisciplinary coordination, technical documentation and delivering structural projects from **concept design through construction**. A motivated and results-driven engineer with **strong attention to detail, a collaborative approach and a positive attitude towards continuous learning**.

CORE COMPETENCIES

- Structural Analysis & Design - Residential, Commercial, Warehouse, Hospital Institutional & Industrial Buildings
- Reinforced Concrete & Steel Design
- Structural Modelling & Finite Element Analysis (ETABS, SpaceGass, SAP2000, SAFE)
- Seismic & Wind Engineering
- Linear and Non-linear Analysis (Response Spectrum Analysis, Pushover Analysis and Time History Analysis)
- Performance-Based Design (PBD)
- Structural Drafting & Drawing (AutoCAD, Revit)
- Site Inspections & Construction Support

TECHNICAL SKILLS

Software: ETABS, SpaceGass, SAP2000, SAFE, RAPT, PERFORM 3D, IDEA StatiCa, CSiBridge, AutoCAD, Revit, VecTor FormWorks, Python, MATLAB, Microsoft Excel

Codes and Standards: Australian Standards (AS 3600, AS 4100, AS 1170), NCC, AASTHO LRFD, ACI 318, AISC 360, ASCE 7, EN 1992, EN 1993, EN 1998, IS 456, IS 1893, NBC, Standards for Performance Based Design

WORK EXPERIENCE

Structural Engineer - Consultant *(Arugatech, Thailand)* **Jan 2024 – Present**

- Performed design and analysis of 10+ reinforced concrete and steel building structures including residential, commercial, warehouse, sports facilities and community centers using ETABS, SpaceGass and SAP2000.
- Prepared detailed engineering calculations, structural drawings, technical reports and design documentation, supporting projects from concept through detailed design and construction.
- Designed reinforced concrete and steel structural elements, retaining walls, foundations and steel connections, including base plates and anchor bolts, using IDEA StatiCa and MS Excel, in accordance with ACI, AISC and other codes, ensuring practical constructability.
- Coordinated with architects, drafters, clients, contractors and other engineers to resolve design issues.
- Conducted site inspections and construction support, reviewing structural works against drawings and design requirements.

Structural Engineer *(Computers and Structures Inc. (CSI) Bangkok)* **Jan 2023 – Dec 2023**

- Validated the analysis, design, and detailing functionality of structural engineering software (CSiBridge-ET, CSiDetail, and CSiCol) through engineering calculations and code-based verification and systematic QA testing.
- Performed independent engineering calculations for concrete bridge components, concrete beams and columns to verify software analysis and design results.
- Designed and verified prestressed concrete (PSC) girders, bent caps, columns and deck using AASHTO LRFD, Caltrans, and Eurocode provisions.
- Developed structural design workflows, engineering logic & calculation algorithms to support software implementation and improve the reliability of structural analysis and design functionality.

- Conducted functional, integration and regression testing of structural engineering software, identifying discrepancies, investigating technical issues and supporting resolution with development teams.
- Prepared technical reports, verification documentation and user manuals and collaborated remotely with international QA and software development teams.

Civil Engineer (Structural Design) ***(TEAM Consultants Pvt. Ltd., Nepal)*** **April 2017 – Oct 2019**

- Performed finite element modelling, linear static and dynamic analysis (Response Spectrum Analysis), calculations, and seismic design of 15+ low to mid-rise reinforced concrete and steel building structures using ETABS & SAP2000.
- Designed and analyzed 2-15 storied RC framed residential buildings, commercial complexes with RC core shear wall system, basement retaining walls & raft slabs and a hospital building extension with an expansion joint.
- Delivered structural drawings, calculation reports and design documentation and technical reports, ensuring compliance with applicable design standards.
- Conducted site inspections to ensure construction quality, verify compliance with structural drawings and specifications, and provide construction support during project delivery.
- Collaborated with drafters, architects, MEP and geotechnical engineers, clients, and contractors to deliver practical, economical and buildable structural solutions.
- Prepared tender documents, cost estimates and BOQ, ensuring compliance with project requirements and specifications.

EDUCATION

Master of Engineering (Structural Engineering) **2020 – 2022**
Asian Institute of Technology, Thailand

Bachelor of Civil Engineering **2011 – 2015**
Tribhuvan University, Nepal

LICENSE AND CERTIFICATE

- **Engineers Australia** - Skills Assessment (Civil Engineer – Structural) - August 2024
- **Nepal Engineering Council** - Registered Professional Engineer - Since June 2016

RESEARCH EXPERIENCE

Master's Thesis - Performance-Based Wind & Seismic Design of a 93-Storey Tall Building **2021-2022**
"The Effect of Performance-Based Wind Design on Seismic Design of a Tall Building in High Seismic and High Wind Zone"
Thesis Advisor: Dr. Naveed Anwar

- 93 Storied building: RCC framed, core wall and BRB provided for lateral force resistance
- Conducted both linear and non-linear wind design and compared the global and local structural responses
- Investigated the influence of performance-based wind design on seismic ductile design, applying performance-based design principles to evaluate structural behaviour under combined high wind and seismic demands.

Student Research Assistant *(Asian Institute of Technology)* **Jan 2021– April 2022**

- Contributed to the development of the textbook "Strut and Tie Method" through literature review and technical writing.
- Conducted research on Advance Concrete Structures and Tall Buildings, and developed academic articles
- Performed software testing and technical documentation for structural engineering software, contributing to the development of CSiDetail & CSi Designer.

Intern Engineer *(HELVETAS Swiss Inter-cooperation, Nepal)* **June 2016– March 2017**

- Performed complete survey of water supply system from the water source to households and assisted in the design of infrastructures such as intake, reservoirs, and pipelines

PUBLICATIONS

- "Moment Curvature Curve: Unlocking the Hidden Treasures"
- "Basic Concepts for Considering Soil-Structure Interaction"